

Space Age

What will a space-faring human race look like? And how will they return back to Earth?

Tim Cook highest-paid US executive

Despite its stock taking a hammering, Apple CEO Tim Cook was recently named the highest-paid chief executive in the US, beating the likes of Larry Ellison and Rupert Murdoch

The medicine of tomorrow

We take a look at some cutting-edge medical innovations which open up a lot of possibilities in the field of medi-care

Spandan Sharma

readersletters@tbinkdigit.com

Medical diagnosis and technology has progressed by great leaps and bounds in recent years. As more and more innovations are discovered, assessing physical and mental health is becoming highly sophisticated. Here are a few of them.

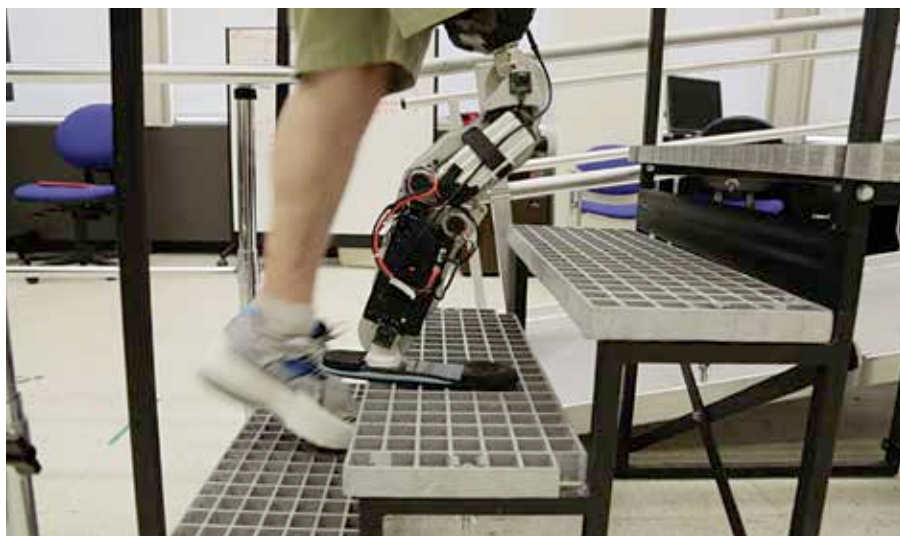
E-Skin sensitive to touch and healing

E-skin, or synthetic skin, is nothing new. Indeed, scientists have been able to create small quantities of fake skin in laboratories for quite some time now, however this material lacks a couple of very important properties human skin possesses. The foremost amongst them is the human skin's sensitivity to temperature and the sense of touch, as well as its ability to heal itself. So while the demand for grafting procedures continues to surge, the possibility of it leaving you a little impaired, and not to mention a little scarred was, until now, very real.

Researchers at Stanford say they have created a synthetic material that has overcome the limitations of traditional synthetic skin. The researchers behind the project say it could lead to smarter pros-



The new E-skin promises to open new frontiers for plastic surgery, despite looking ugly



Zac Vawter tests out the new bionic leg design... Remind anyone else of the Terminator

thetics, more resilient and self-repairing electronics and, of course, better synthetic skin. The team started the project with an objective of finding a material that could heal itself, as well as conduct electricity. They found that self-healing was possible if the material was made of long chains of molecules, connected by hydrogen bonds that allowed weak attraction between differently charged regions of atoms. The material is highly flexible as the molecules can split up easily, yet come back together in their original structure.

Material strength can be increased by embedding metals like nickel, that also allows electric charges to flow easily from one particle to another. The final result is a material sensitive enough to detect the most minor of pressures as well as the slightest change in temperature. All hail, the future of skin is here!

Thought-controlled bionic leg

Prosthetic limbs have seen steady progress in the quality of materials used in them, with lighter and quicker devices made from new-age materials. However, the biggest problem that remains is their

usability. Apart from prosthetic limbs being bulky, they are difficult to control. Now, researchers in Chicago have come up with a device that may just be the solution to this problem.

Rehabilitation Institute of Chicago (RIC) researchers have created a bionic leg that responds to electrical impulses from muscles in the patient's hamstring, with the user's thoughts synchronising the movements of the prosthetic ankle and knee. The device was tested by amputee Zac Vawter who used it to climb to the top of the Chicago Willis Tower (that's 103 flights of stairs!). He completed the climb in just under an hour, with no breaks, and has successfully proved the usability of the device.

Vawter, who lost his right leg in a motorcycle accident in 2009, underwent a surgical procedure called "targeted muscle reinnervation". The procedure reassigned nerves, allowing him to have more natural control of the prosthetic device. RIC software engineers, physical therapists and doctors worked in a coordinated effort to refine the bionic leg's controls, that read the user's "intent" and work together. The